Arjuna JEE 2027

JEE MAINS & IIT JEE ADVANCED



MATHEMATICS

BASIC MATHEMATICS

DIBY (DO IT BY YOURSELF)-(61 to 122)

By: Sachin Jakhar (Defender Mamu)

DIBY-61

Number of solution of the equation $3^{\log_3^2 x} + x^{\log_3 x} = 162$ is:

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

[Ans. : (C)]

DIBY-62

Solve: $\log_{10} 5 + \log_{10}(x+10) - 1 = \log_{10}(21x-20) - \log_{10}(2x-1)$

[Ans. : 3/2, 10]

DIBY-63

Solve: $\log_6 2^{x+3} - \log_6(3^x - 2) = x$

[Ans. : $\log_3 4$]

DIBY-64

Solve: $\log_{0.5x} x^2 - 14\log_{16x} x^3 + 40\log_{4x} \sqrt{x} = 0$

[Ans. : $\frac{1}{\sqrt{2}}, 1, 4$]

DIBY-65 (Acidic)

Solve: $2\log_2\log_2 x + \log_{1/2}\log_2\left(2\sqrt{2}x\right) = 1$

[Ans. : 8]

DIBY-66 (Challenger)

Solve: $\log_{x^2+6x+8}\left(\log_{2x^2+2x+3}\left(x^2-2x\right)\right) = 0$

[Ans. : –1]

DIBY-67

If x_1 and x_2 are the roots of equation $e^{3/2} \cdot x^{2\ln x} = x^4$, then the product of the roots of the equation is

(A) e^2 (B) e (C) $e^{3/2}$ (D) e^{-2}

[Ans. : (A)]

DIBY-68

The solution set of inequality: $\log_{(3x^2+1)} 2 < \frac{1}{2}$

[Ans. : $(-\infty, -1) \cup (1, \infty)$]

DIBY-69

The number of positive integers not satisfying the inequality

$\log_2(4^x - 2 \cdot 2^x + 17) > 5$.

(A) 2 (B) 3 (C) 4 (D) 1

[Ans. : (A)]

DIBY-70

	Column-I		Column-II
A.	$5^x - 24 = \frac{25}{5^x}$	p.	-3
B.	$\left(2^{x+1}\right)(5^x) = 200$	q.	-2
C.	$4^{2/x} - 5\left(4^{1/x}\right) + 4 = 0$	r.	-1
D.	$2^{2x+1} - 33\left(2^{x-1}\right) + 4 = 0$	s.	0
E.	$\frac{2^{x-1} \cdot 4^{x+1}}{8^{x-1}} = 16$	t.	1
F.	$3^{2x+1} + 10(3^x) + 3 = 0$	u.	2
G.	$64(9^x) - 84(12^x) + 27(16^x) = 0$	v.	3
H.	$5^{2x} - 7^x - 5^{2x}(35) + 7^x(35) = 0$	w.	None

Ans. :

A → u;

B → u;

C → t;

D → q, v;

E → p, q, r, s, t, u, v;

F → w;

G → t, u;

H → s

QUESTIONS ON MODULUS

DIBY-71

The value of x satisfying the equation $x + |3x - 2| = 2$

(A) $x = \{0, 1\}$ **(B)** $x = \left\{0, \frac{2}{3}\right\}$ **(C)** $x = \{1, 2\}$ **(D)** $x = \left\{1, \frac{4}{3}\right\}$

[Ans. : (A)]

DIBY-72

The value of x satisfying the equation $|\log_2 x| = 3$

(A) $x = \{8, -8\}$ **(B)** $x = \left\{8, \frac{1}{8}\right\}$ **(C)** $x = \left\{6, \frac{1}{6}\right\}$ **(D)** $x = \left\{-8, -\frac{1}{8}\right\}$

[Ans. : (B)]

DIBY-73

The value of x satisfying the equation $|3x - 2| + x = 11$

(A) $x = \left\{\frac{13}{4}, -\frac{9}{2}\right\}$ **(B)** $x = \left\{\frac{13}{4}, \frac{9}{2}\right\}$ **(C)** $x = \left\{-\frac{13}{4}, -\frac{9}{2}\right\}$ **(D)** $x = \left\{\frac{13}{4}\right\}$

[Ans. : (A)]

QUESTIONS ON MODULUS

DIBY-74

The value of x satisfying the equation $|x-3|+2|x+1|=4$

(A) 1 (B) -1 (C) $\frac{5}{3}$ (D) 3

[Ans. : (B)]

DIBY-75

The number of real solution of the equation, $x^2 - |x| - 12 = 0$ is [JEE Mains-2021]

(A) 2 (B) 3 (C) 1 (D) 4

[Ans. : (A)]

DIBY-76

Solve $|x^2 + 4x + 3| + 2x + 5 = 0$ [JEE-1987, 5M]

[Ans. : $x = -4$ and $(-1-\sqrt{3})$]

DIBY-77

The sum of all the real roots of the equation $|x-2|^2 + |x-2| - 2 = 0$ is _____. [JEE-1997, 2M]

[Ans. : 4]

QUESTIONS ON MODULUS

DIBY-78

$(x-3)^2 + |x-3| - 11 = 0$, the sum of solutions of this equation is?

[JEE Mains-2021]

(A) 5 (B) 6 (C) 7 (D) 8

[Ans. : (B)]

DIBY-79

The sum of all roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is

[JEE Mains-2023]

(A) $9 - \sqrt{3}$ (B) $9 + \sqrt{3}$ (C) $11 - \sqrt{3}$ (D) $11 + \sqrt{3}$

[Ans. : (B)]

DIBY-80

The complete set of value of x satisfying the equation $|x| - |x-2| = 2$

(A) $x \in (-\infty, 0]$ (B) $x \in [0, 2]$ (C) $x \in [2, \infty)$ (D) $x \in \{0, 2\}$

[Ans. : (B)]

QUESTIONS ON MODULUS

DIBY-81

The number of real roots of the equation $|x|^2 - 3|x| + 2 = 0$ is

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

[Ans. : (D)]

DIBY-82

Solutions of $|4x + 3| + |3x - 4| = 12$ are

(A) $x = -\frac{7}{3}, \frac{3}{7}$ **(B)** $x = -\frac{5}{2}, \frac{2}{5}$ **(C)** $x = -\frac{11}{7}, \frac{13}{7}$ **(D)** $x = -\frac{3}{7}, \frac{7}{5}$

[Ans. : (C)]

DIBY-83

Solve if $|x - 5| + |x + 4| = 9$

(A) $[-4, 5]$ **(B)** $(-4, 5)$ **(C)** $(-4, 5]$ **(D)** $[-4, 5)$

[Ans. : (A)]

QUESTIONS ON MODULUS

DIBY-84

Number of real solutions of $|x-1| = |x-2| + |x-3|$ is

(A) 0 (B) 1 (C) 2 (D) more than 2

[Ans. : (C)]

DIBY-85

Solve for x :

(i) $|||x+5|-3|-1| = 2$

(ii) $||||x-5|-7|-3|-2| = 1$

[Ans. : (i) $x \in \{-11, -5, -1\}$
(ii) $x \in \{-8, -6, -2, -4, 0, 2, 4, 6, 8, 10, 12, 14, 16, 18\}$]

DIBY-86

Solution of Modulus equation: $|x-3| + 2|x+1| = 4$

[Ans. : (x = –1)]

DIBY-87

Solve for x :

(i) $\frac{1}{2} \leq |2x-1| \leq \frac{3}{5}$

(ii) $-\frac{1}{3} \leq |3x-4| \leq 2$

(iii) $2 \leq |4-5x| \leq \frac{10}{3}$

(iv) $3 < |x^2-1| < 8$

Ans. :

(i) $x \in \left[\frac{1}{5}, \frac{1}{4}\right] \cup \left[\frac{3}{4}, \frac{4}{5}\right]$

(ii) $x \in \left[\frac{2}{3}, 2\right]$

(iii) $x \in \left[\frac{2}{15}, \frac{2}{5}\right] \cup \left[\frac{6}{5}, \frac{22}{15}\right]$

(iv) $x \in (-3,-2) \cup (2,3)$

DIBY-88

The complete set of real 'x' satisfying $||x-1|-1| \leq 1$ is

(A) [0, 2] **(B)** [–1, 3] **(C)** [–1, 1] **(D)** [1, 3]

[Ans. : (B)]

DIBY-89

Solve : $\left|\frac{x-3}{x+1}\right| \leq 1$

[Ans. : $x \geq 1$**]**

DIBY-90

Solve : $\left|1+\frac{3}{x}\right| > 2$

[Ans. : $x \in (-1, 0) \cup (0, 3)$**]**

DIBY-91

Number of non-positive integral values of 'x' satisfying the given inequality, $\left|x^2-1\right| \leq |2x-1|$ **is**

(A) 0 (B) 1 (C) 2 (D) 3

[Ans. : (D)]

DIBY-92

Solve for x :

(i) $\frac{(|x|-1)}{(|x|-2)} \leq 0$

(ii) $\frac{(|x|-1)(|x|-3)}{|x|^2-2|x|} \geq 0$

(iii) $\frac{(|x|^2-5|x|+6)}{(4-|x|^2)} \leq 0$

[Ans. :

(i) $x \in (-2,-1] \cup [1,2)$

(ii) $x \in (-\infty,-3] \cup (-2,-1] \cup [3,\infty) \cup [1,2)$

(iii) $x \in (-\infty,-3] \cup [3,\infty)$**]**

QUESTIONS ON MODULUS

DIBY-93

Solve : $\frac{|x-2|-3}{|x+5|+2} \geq 0$.

[Ans. : $x \in (-\infty, -1] \cup [5, \infty)$ **]**

DIBY-94

Solve : $\left|\frac{3}{x-5}\right| > 1$.

[Ans. : $x \in (2, 8) - \{5\}$**]**

DIBY-95

Solve : $|x^2 - 3x - 3| > |x^2 + 7x - 13|$.

[Ans. : $x \in (-\infty, -4) \cup (1, 2)$**]**

DIBY-96

The number of integral solutions of $|\log_5 x^2 - 4| = 2 + |\log_5 x - 3|$ is

(A) 1 (B) 2 (C) 3 (D) 0

[Ans. : (A)]

DIBY-97

Value of x satisfying: $\log_4(x^2 - 1) - \log_4(x - 1)^2 = \log_4|4 - x|$ is $a + \sqrt{2a}$, then value of a is _____.

[Ans. : 03]

DIBY-98

Exhaustive set of values of x satisfying $\log_{|x|}(x^2 + x + 1) \geq 0$ is

(A) $(-1, 0)$
(B) $(-\infty, -1) \cup (1, \infty)$
(C) $(-\infty, \infty) - \{-1, 0, 1\}$
(D) $(-\infty, -1) \cup (-1, 0) \cup (1, \infty)$

[Ans. : (D)]

DIBY-99

Number of values of x satisfying the equation $x^4 = |x|^{\log_2(x^2+12)}$ is

(A) 2 (B) 3 (C) 4 (D) 5

[Ans. : (D)]

DIBY-100

The inequality $|\log_e|x|| < 1$ is satisfied if

(A) $x \in (1/e, e)$ (B) $x \in (-e, -1/e)$ (C) $x \in (-1, 1)$ (D) $x \in (-e, e)$

[Ans. : A, B]

DIBY-101

Find the number of integral solution of the equation

$\log_{\sqrt{x}}(x + |x-2|) = \log_x(5x - 6 + 5|x-2|)$.

[Ans. : 1]

QUESTIONS ON SPECIAL PROERTY OF MOD

DIBY-102

Solve: $|x^2 - 4x + 3| + |x^2 - 11x + 24| = 7|x - 3|$.

[**Ans.** : $x \in [1, 8]$]

DIBY-103

Solve: $|x^2 - 3x + 2| + |x + 5| > |x^2 - 2x + 7|$.

[**Ans.** : $x \in (-\infty, -5) \cup (1, 2)$]

DIBY-104

Solve x : $|x^2 + 5x + 9| < |x^2 + 2x + 2| + |3x + 7|$.

[**Ans.** : $x < -\frac{7}{3}$]

DIBY-105

The complete set of real values of x satisfying the inequality $\left(2\log_3^2 x - 3\log_3 x - 8\right)\left(2\log_3^2 x - 3\log_3 x - 6\right) \geq 3$ is

[Ans. : $x \in \left(0, \frac{1}{3\sqrt{3}}\right] \cup \left[\frac{1}{3}, 9\sqrt{3}\right] \cup [27, \infty)$]

DIBY-106

The complete set of real values of x satisfying the inequality $\log_3 \frac{|x^2-4x|+3}{x^2+|x-5|} \geq 0$ is

(A) $\left(-\infty, \frac{1}{2}\right]$

(B) $\left(-\infty, \frac{2}{5}\right] \cup [2, \infty)$

(C) $\left(-\infty, \frac{2}{5}\right] \cup [1, \infty)$

(D) $\left(-\infty, \frac{1}{2}\right] \cup [1, \infty)$

[Ans. : (D)]

DIBY-107

If $|x-2|+2|x+2|=P$, then find P is given equation has

(i) no solution
(ii) 1 solution
(iii) 2 solutions
(iv) more than 2 solutions

Ans. :

(i) No solution $P<4$
(ii) 1 solution $P=4$
(iii) 2 solutions $P>4$
(iv) More then 2 solution $P\in\phi$

DIBY-108

Find the value of k if equation $|x+2|-2|x-7|+3|x-4|=k$ has

(i) one solution (ii) two solutions
(iii) 3 solutions (iv) 4 solutions
(v) no solution (vi) ∞ solutions

Ans. :

(i) one solution $k\in\phi$
(ii) 2 solutions $k>0$
(iii) 3 solutions $k\in\phi$
(iv) 4 solution $k\in\phi$
(v) no solution $k<0$
(vi) ∞ solutions $k=0$

DIBY-109

Draw graph of $y = ||x - 2| - 3|$ and answer following questions:

(i) Minimum value of y is _____.

(ii) If $y = k$ has 2 solutions then k is _____.

(iii) If $y = k$ has 3 solutions then k is _____.

(iv) If $y = k$ has 4 solutions then k is _____.

(v) If $y = k$ has no solution then k is _____.

(vi) Number of solutions of : $||x - 2| - 3| = \frac{2x+10}{15}$, is(are) _____.

Ans. :

(i) 0

(ii) $k \in (3, \infty) \cup \{0\}$

(iii) $k = 3$

(iv) $k \in (0, 3)$

(v) $k \in (-\infty, 0)$

(vi) 4

QUESTIONS ON IRRATIONAL EQUATIONS & INEQUALITY

DIBY-110

It x is a real number, find the number of integral values of x for which the expression $\sqrt{|x-2|-3}+\sqrt{5-|x+1|}$ is defined.

[Ans. : 6]

DIBY-111

If the complete solution set of the inequality $1-\sqrt{1-4x^2}<3x$ is $(a,b]$, find the value of $10a+2b$.

[Ans. : 1]

DIBY-112

Solve : $\sqrt{x-2}\geq\sqrt{5-x}$

[Ans. : $x\in\left[\frac{7}{2},5\right]$]

DIBY-113 (Challenger)

Solve : $\sqrt{x-6}-\sqrt{10-x}\geq 1$

[Ans. : $x\in\left[8+\frac{\sqrt{7}}{2},10\right]$]

DIBY-114

Which of the following is true?

(A) $(\log_{10} 2)^2 + 1 > \log_{10} 4$

(B) $\log_{10} 90 > \log_5 50$

(C) $\log_4 \log_3 \log_2 16 > \log_{16} 4$

(D) $2(\log_{10} 3)^2 - 3(\log_{10} 2)^2 > (\log_{10} 2) \times (\log_{10} 3)$

[Ans. : A, D]

DIBY-115 (Challenger)

Positive numbers x, y **and** z **satisfy** $xyz = 10^{81}$ **and**
$(\log_{10} x)(\log_{10} yz) + (\log_{10} y)(\log_{10} z) = 468.$
Find the value of $(\log_{10} x)^2 + (\log_{10} y)^2 + (\log_{10} z)^2$

[Ans. : 5625]

DIBY-116

Consider the system of equations $\log_{10}(2000xy) - \log_{10} x \cdot \log_{10} y = 4,$
$\log_{10}(2yz) - \log_{10} y \cdot \log_{10} z = 1$ **and** $\log_{10}(zx) - \log_{10} z \cdot \log_{10} x = 0,$ $x, y, z \leq 90$
then $x + y + z =$

[Ans. : 7]

DIBY-117

Let $S = \left\{x \in R: \left(\sqrt{3}+\sqrt{2}\right)^x + \left(\sqrt{3}-\sqrt{2}\right)^x = 10\right\}$. Then the number of elements in S is

[JEE Mains-2024]

(A) 4 (B) 0 (C) 2 (D) 1

[Ans. : (C)]

DIBY-118

The number of elements in the set $\{n \in \mathbb{Z}: |n^2 - 10n + 19| < 6\}$ is

[JEE Mains-2023]

[Ans. : 6]

DIBY-119

Let $S = \left\{x \in [-6,3] - \{-2,2\}: \frac{|x+3|-1}{|x|-2} \geq 0\right\}$ and $T = \{x \in Z: x^2 - 7|x| + 9 \leq 0\}$. Then the number of elements of $S \cap T$ is

[JEE Mains-2022]

(A) 7 (B) 5 (C) 4 (D) 3

[Ans. : (D)]

DIBY-120

Let $A = \{x: |x^2 - 10| \leq 6\}$ and $B = \{x: |x - 2| > 1\}$. Then [JEE Mains-2026]

(A) $A - B = [2, 3)$ **(B)** $A \cup B = (-\infty, 1] \cup (2, \infty)$

(C) $B - A = (-\infty, -4) \cup (-2, 1) \cup (4, \infty)$ **(D)** $A \cap B = [-4, -2] \cup [3, 4]$

[Ans. : (C)]

DIBY-121

If $A = \{x \in R: |x - 2| > 1\}, B = \{x \in R: \sqrt{x^2 - 3} > 1\}, C = \{x \in R: |x - 4| \geq 2\}$ and Z is the set of all integers, then the number of subsets of the set $(A \cap B \cap C)^C \cap Z$ is

[JEE Mains-2021]

[Ans. : 256]

DIBY-122

The number of elements in the set $\{x \in R: (|x| - 3)|x + 4| = 6\}$ is equal to

[JEE Mains-2021]

(A) 3 **(B)** 2 **(C)** 4 **(D)** 1

[Ans. : (B)]

It is never about end result

It's always about Journey

THANK You

#futureIITians